

ITCS231: Data Structure and Algorithm

Faculty of ICT, Mahidol University

Project 3: Sorting with the Generic datatype

So far, we have learnt many sorting algorithms, both basic and advanced. To ensure that you really understand how to implement these algorithms, you will have to extend what you have learnt in class by implementing the following method.

- `static <T> void insertionSort(T[] a, Comparator<T> c)` (3 point)
- `static <T> void quickSort(T[] a, Comparator<T> c)` (3 point)
- `static <T> void mergeSort(T[] a, Comparator<T> c)` (4 point)

Please follow the instruction below to start and submit the project:

- Download starter file “MySorting.java” from MyCourses.
- Write down your student ID and Name at the top of the file.
- Implement three sorting algorithms above with Generic Type T and Comparator.
- Submit by uploading it to MyCourses. Please .zip the file as your student ID e.g., [6488XXX.zip](#)
- *Failure to follow any of these instruction, your work will not be graded.

Note that: The Comparator<T> interface is helping on how to compare object of type T. The method `compare(T a, T b)` will return integer with meaning as follows:

- if $a < b$, returns **a negative value**;
- if $a == b$, returns 0;
- if $a > b$, returns **a positive value**.

You should submit this project on the MyCourses system by **26 November 2021, 11.55 p.m.**